

Exploratory Data Analysis (EDA) Summary Report

1. Introduction

The purpose of this report is to explore the Geldium delinquency prediction dataset, identify important data quality issues, understand patterns related to customer delinquency, and highlight potential risk indicators. The analysis focuses on missing values, inconsistent data, anomalies, and relationships between customer financial characteristics and delinquency status.

2. Dataset Overview

The dataset contains customer financial, credit, employment, and payment-related information that can be used to analyze factors associated with delinquency.

Core Dimensions

- Number of records:** 500
- Number of variables:** 19
- Target variable:** Delinquent_Account
- Identifier:** Customer_ID

Variable Types

- Numerical:** Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Loan_Balance, Debt_to_Income_Ratio, Account_Tenure
- Categorical:** Employment_Status, Credit_Card_Type, Location, Month_1 to Month_6

The dataset contains **80 delinquent customers (16%)** and **420 non-delinquent customers (84%)**, showing an imbalance in the target variable. During the initial review, missing values were identified in Income, Loan_Balance, and Credit_Score. Inconsistent labels were found in Employment_Status, including EMP, employed, and Employed. Four records had Credit_Utilization values above 100%, which require further investigation. No duplicate records or duplicate Customer_ID values were identified.

3. Missing Data Analysis

Identifying and addressing missing data is important for maintaining data quality and retaining useful customer information.

- Variables with missing values:** Income (39 missing, 7.8%), Loan_Balance (29 missing, 5.8%), Credit_Score (2 missing, 0.4%), and Employment_Status (inconsistent category labels).

Missing data treatment:

Data Issue	Treatment	Justification
Income (7.8%)	Median imputation	Retains records and reduces the effect of extreme income values.
Loan_Balance (5.8%)	Median imputation	Provides a robust estimate without removing customer records.
Credit_Score (0.4%)	Median imputation	Only a small number of values are missing.
Employment_Status (labels)	Category standardization	EMP, employed, and Employed were standardized to Employed.
Credit_Utilization (>100%)	Retained for investigation	These are anomalies rather than missing values and require business validation.

Median imputation was selected because the missing percentages were relatively small and the approach allows existing customer records to be retained. Synthetic data generation was not considered necessary.

4. Key Findings and Risk Indicators

The analysis identified trends and patterns that may indicate risk factors for delinquency. Feature relationships and statistical correlations were explored to uncover insights relevant to predictive modeling.

- **Correlations observed between key variables:** Delinquent customers showed a higher proportion of Late payments in Month 4, indicating recent payment behavior is an early warning signal. Average credit utilization was 0.507 for delinquents vs 0.489 for non-delinquents. Debt-to-income ratio averaged 0.306 for delinquents vs 0.297 for non-delinquents. Credit score requires further distribution analysis. Missed payments showed an unexpected average of 2.85 for delinquents vs 2.99 for non-delinquents, requiring deeper review.
- **Unexpected anomalies:** Four records have Credit_Utilization values above 100%, Employment_Status contains inconsistent category labels, and the average missed-payment pattern deviates from expectations.

5. AI & GenAI Usage

Generative AI tools were used to summarize the dataset, impute missing data, and detect patterns. This section documents AI-generated insights and the prompts used to obtain results.

Example AI prompts used:

- *'Summarize key patterns in the dataset and identify anomalies.'*
- *'Suggest an imputation strategy for missing income values based on industry best practices.'*
- *'Identify missing, inconsistent, or anomalous data that could affect delinquency analysis.'*
- *'Analyze the relationship between the available variables and delinquency status.'*

AI-generated suggestions were reviewed against the actual dataset before being included in the analysis.

6. Conclusion & Next Steps

The EDA successfully identified and treated critical data quality issues, including missing values using median imputation and inconsistent labels through standardization. The analysis highlights that recent payment behavior, credit utilization, and debt-to-income ratio are valuable signals associated with customer delinquency. Next steps include further investigating the unexpected missed-payment pattern and building predictive models incorporating these risk indicators.